MODULE 1 (CONTINUED): SOUND MATH & COMPRESSION ALGORITHMS (TB PG 38–44)

1.3.3 Sound File Size Calculations (TB Pg 38–40)

Sound is analogue by nature. To store it digitally, sound waves are sampled at regular intervals:

- **Sample Rate:** Number of sound samples taken per second, measured in Hertz (Hz) or kHz (e.g. CD quality = 44,100 Hz).
- **Sample Resolution:** Number of bits allocated per sample (bit depth, e.g. 16 bits).
- **Duration:** Total time of audio in seconds.
- **Channels:** Mono = 1 channel; Stereo = 2 channels.

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Formula: Sound File Size (bits) = Sample Rate (Hz) × Sample Resolution (bits) × Duration (s) × Channels
In Bytes: File Size (Bytes) = File Size (bits) ÷ 8
In Mebibytes (MiB): File Size (MiB) = File Size (Bytes) ÷ (1024 × 1024)
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Worked Example: A 30-second stereo audio track sampled at 44.1 kHz (44,100 Hz) with 16-bit resolution:
1. Bits = 44,100 × 16 × 30 × 2 = 42,336,000 bits.
2. Bytes = 42,336,000 ÷ 8 = 5,292,000 Bytes.
3. MiB = 5,292,000 ÷ 1,048,576 ≈ **5.05 MiB**.

1.3.4 Lossy vs Lossless Compression (TB Pg 41–44)

Feature	Lossy Compression	Lossless Compression
Data Removal	Permanently removes redundant or imperceptible data (e.g. sounds human ears cannot hear).	No data is lost. Original file can be decompressed back to 100% exact original form.
File Size Reduction	Massive reduction (up to 90% smaller).	Moderate reduction (usually 30% to 50% smaller).
Common Formats	JPEG (images), MP3/AAC (audio), MP4 (video).	PNG, SVG (images), FLAC, ALAC (audio), ZIP, TXT.
Suitable Uses	Web streaming, photos where slight quality drop is acceptable.	Text files, executable programs, spreadsheets, medical scans.

1.3.5 Run-Length Encoding (RLE) Algorithm (TB Pg 43–44)

Concept: RLE reduces the size of repeated sequences of identical data (runs). It stores the **count (run length)** followed by the **value**.
• Uncompressed Text: `AAAAABBBCCCCCDD` (15 bytes)